Famicom JAMMA board

power

File: power.kicad_sch

cpu

File: cpu.kicad_sch

ppu

File: ppu.kicad_sch

cart

File: cart.kicad_sch

Power: +5V entry, switched +5V_SW, +3V3, supply of the RP2A03 module

CPU: RP2A03 module, RAM, address decoder, data-bus resistors

PPU: RP2C02, address latch, video RAM, sprite detection

Cartridge connector and the dump circuit for emulation mode

video

File: video.kicad_sch

audio

File: audio.kicad_sch

jamma

File: jamma.kicad_sch

mcu

File: mcu.kicad_sch

Video: RGB resistor DAC and composite sync

Audio: mixer, loop through the cartridge, power amplifier

JAMMA edge connector, inputs, coin counter and lockout drivers

Three Raspberry Pi Pico 2 modules, USB Type-C socket, links

PWR_FLAG

GND

PWR_FLAG

+5V

PWR_FLAG

VSYS

Sheet: /
File: fcjamma.kicad_sch

Title: Famicom JAMMA board

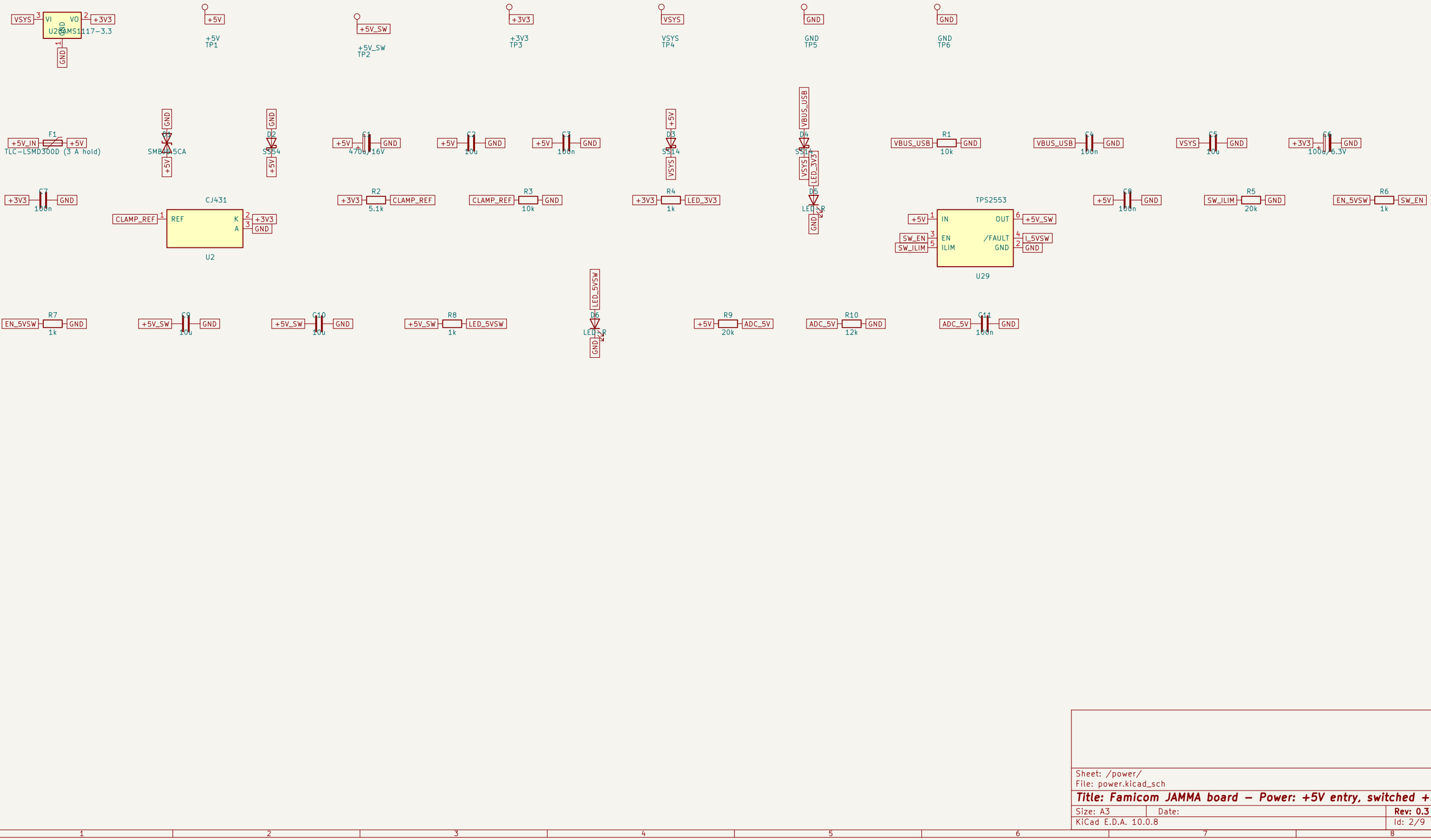
Size: A3	Date:	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 1/9

Power: +5V entry, switched +5V_SW, +3V3, supply of the Pico modules

+5V_IN comes from the JAMMA fingers 3, 4, C, D. −5 V and +12 V are not used. Supply range at the fingers: 4.75 V to 5.25 V.

+5V_SW feeds the CPU, the PPU, the memories, the 5 V logic and the cartridge through the power switch U29. It is off until Pico C raises EN_5VSW, so that no 5 V signal reaches a Pico pin before that Pico is powered (the RP2350 pins tolerate 5 V only while powered). U29 limits the current to about 1.3 A: a short circuit in a cartridge cannot draw the cabinet supply's full current.

VSYS feeds the three Pico modules and the +3V3 regulator from +5V or from the USB socket, whichever is present.



Sheet: /power/
File: power.kicad_sch

Title: Famicom JAMMA board – Power: +5V entry, switched +5V_SW, +3V3, s

Size: A3	Date:	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 2/9

CPU: RP2A03, work RAM, address decoder, data-bus resistors

U1 is the donor's RP2A03, plugged into a 40-pin socket. In emulation mode the socket is empty and the dump shift registers drive the buses.

The 470 ohm resistors let Pico B replace what the PPU sees on a write without disturbing the CPU side of the bus.

No pull-ups on the CPU data bus: they are known to upset some PPU revisions, and games read the floating bus.

The logic next to the CPU and the PPU is 74HCT: its inputs switch at TTL levels, which is what the two NMOS chips deliver.

RP2A03 (donor)

U1

JP3

CLK_SEL

RN1

PPU_RST_N

RN2

IRQ_N

NMI_N

OE1_N

OE2_N

IS62C256AL (work RAM)

U3

RN3

WRAM_A11

WRAM_A12

WRAM_A13

WRAM_A14

74HCT138

U4

74HCT00

U5

RN6

RN7

CPU bus (option)

J3

M2

TP7

CLK

TP8

CLK_SEL

TP9

C12 100n

C13 100n 16V

R11 33

R12 470

R13 470

R14 47k

R15 47k

R16 47k

R17 47k

R18 47k

R19 47k

R20 47k

C14 100n

C15 100n

R21 33

C16 100n

R22 470

R23 470

R24 470

R25 470

R26 470

R27 470

R28 470

R29 470

R30 100

R31 100

R32 100

R33 100

R34 100

R35 100

R36 100

R37 100

R38 100

R39 100

R40 100

R41 100

R42 100

R43 100

R44 100

R45 100

RN4

RN5

R46 10k

R47 12k

RAM_A11-A14 (option)

J4

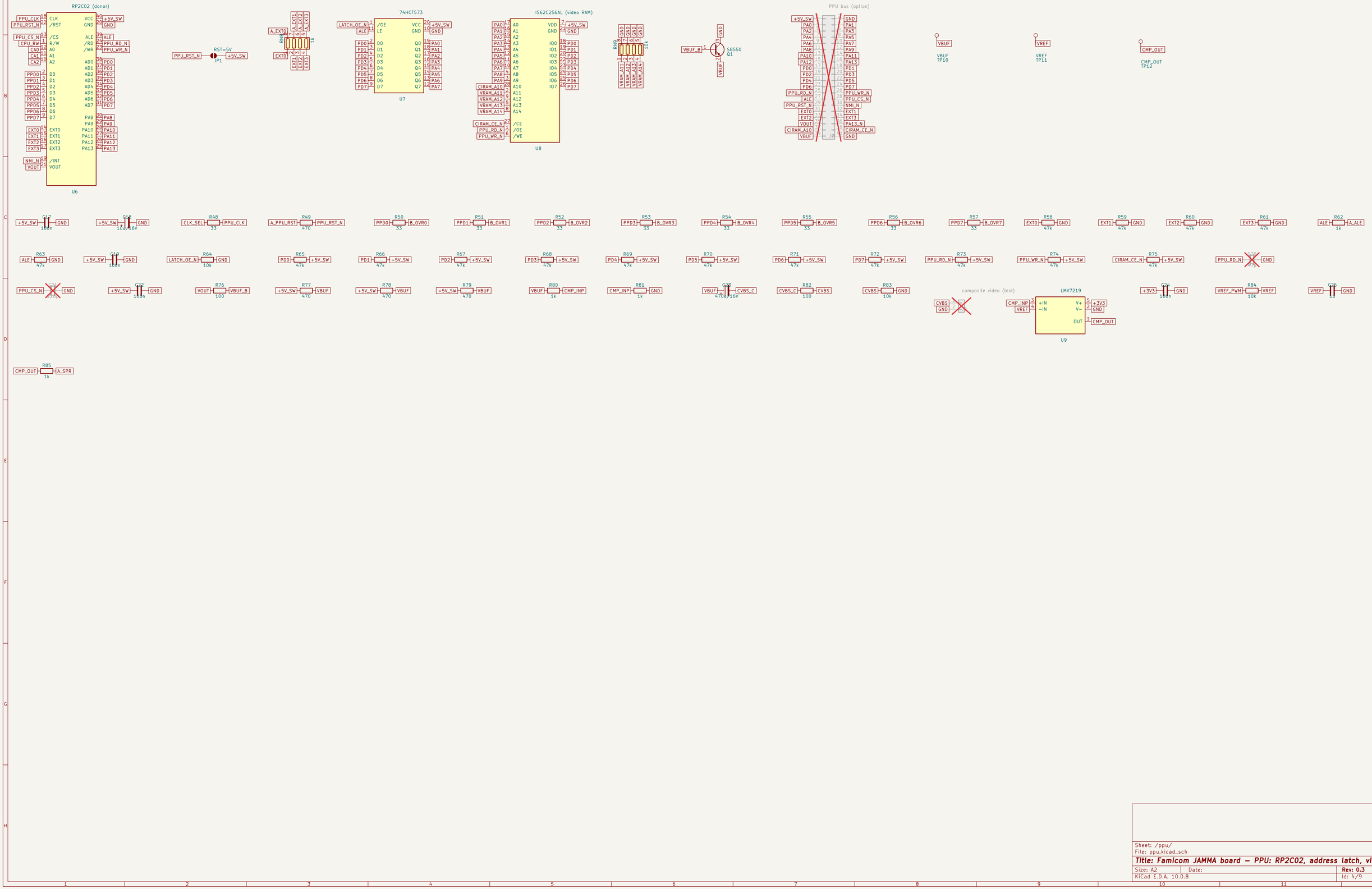
PPU: RP2C02, address latch, video RAM, sprite detector

U6 is the donor's RP2C02, plugged into a 40-pin socket. EXT0–3 give the lower four bits of the palette index of every pixel once

Pico B has forced bit 6 of register 2000 high. The fifth bit (sprite or background) is read from the brightness of VOUT:

Pico B makes the PPU store a dark colour in every background entry and a bright one in every sprite entry.

EXT shows entry 0 of the current background palette (4, 8 or 12) on transparent pixels, and 0 while rendering is off: the firmware maps these.



Cartridge connector and the dump circuit for emulation mode

J8 is a new 60-pin card-edge connector. Pin 45 carries the board's sound to the cartridge, pin 46 brings it back.

Dump circuit: with the CPU and PPU sockets empty, Pico C reads the cartridge through shift registers and hands the data to Pico A.

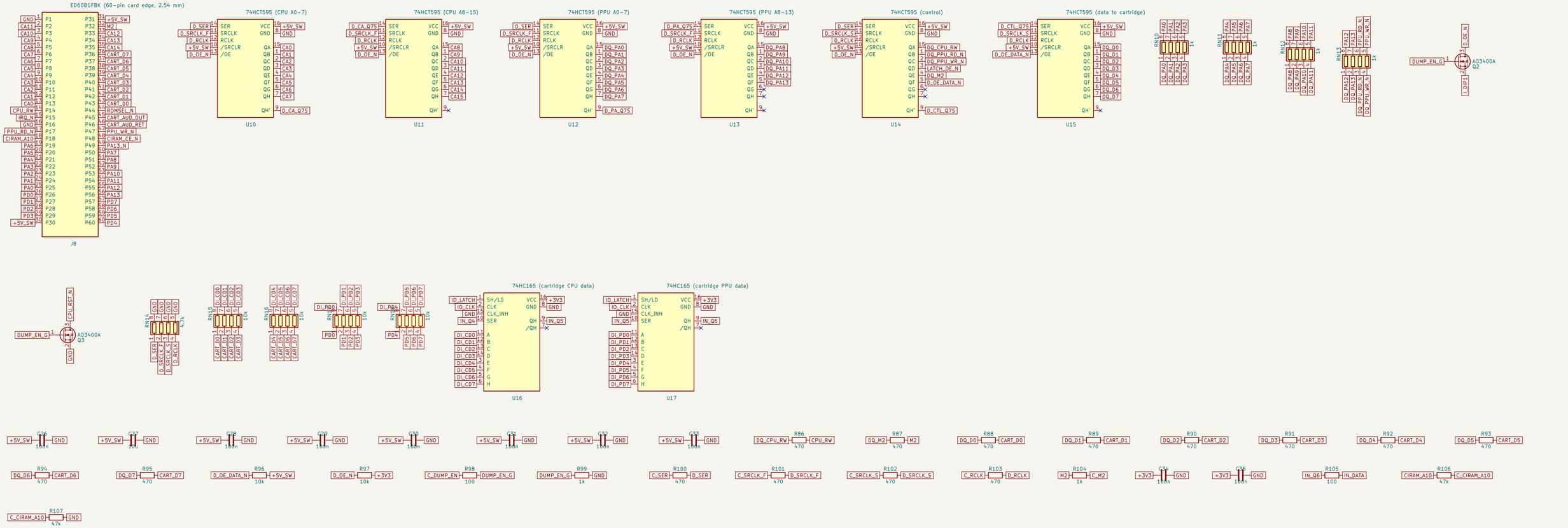
The six 74HCT595 run from +5V_SW and take the 3.3 V signals of Pico C through 470 ohm.

Their outputs can be switched on only while DIP switch 1 is set to emulation AND Pico C raises C_DUMP_EN (Q with its source on the switch).

C_DUMP_EN also holds the CPU in reset (all its bus pins are released then). A PPU cannot be switched off its bus, so every driver that

meets a PPU output has 1 k in series: a PPU left in its socket by mistake sees at most 5 mA per pin. The firmware checks for chips first.

Pico C must hold its four lines to the 74HCT595 low whenever +5V_SW is off.

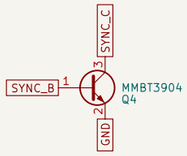


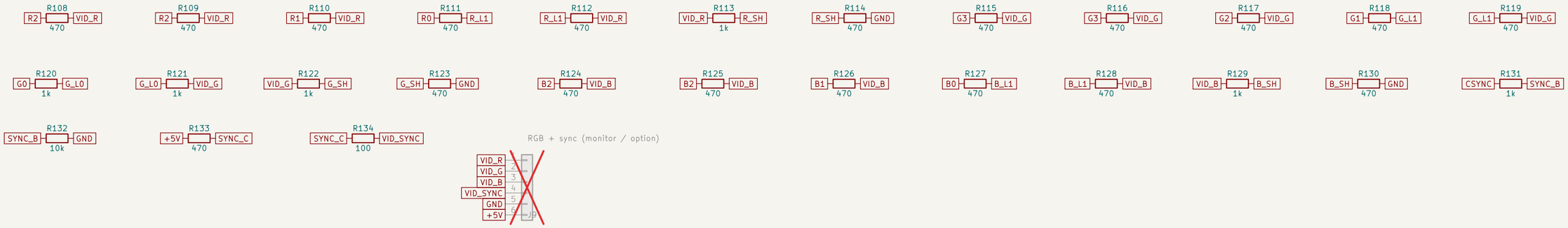
Sheet: /cart/	
File: cart.kicad_sch	
Title: Famicom JAMMA board – Cartridge connector and the dump circuit for	
Size: A2	Date:
KiCad E.D.A. 10.0.8	**Rev: 0.3**
	Id: 5/9

Video: RGB resistor DAC and composite sync

Pico A drives the DAC directly. Full scale is 3.0 V open circuit, about 2.85 V into 2 k, source resistance about 125 ohm.

Red and blue have 3 bits, green has 4. Only 470 ohm and 1 k are used so that the parts list stays short.





RGB + sync (monitor / option)

Sheet: /video/
File: video.kicad_sch

Title: Famicom JAMMA board – Video: RGB resistor DAC and composite sync

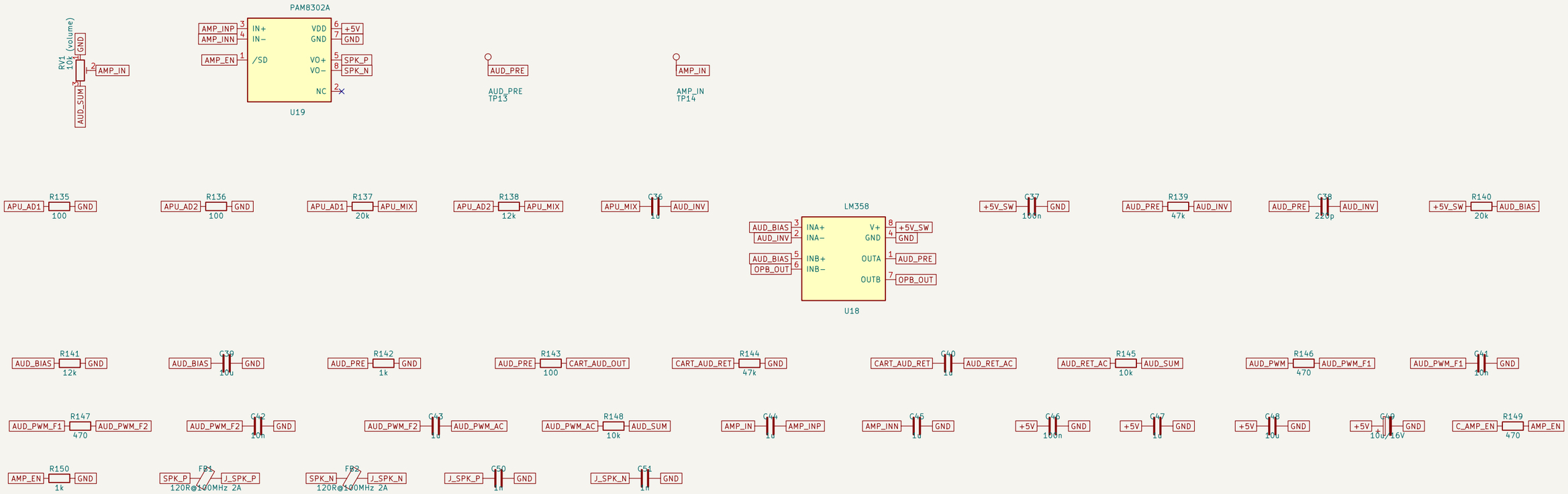
Size: A3	Date:	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 6/9

Audio: mixer, loop through the cartridge, power amplifier

Real-chip mode: 2A03 pins -> mixer (Famicom values) -> cartridge pin 45 -> cartridge -> pin 46 -> volume -> amplifier.

Emulation mode: Pico A's PWM output -> two RC stages -> volume -> amplifier.

The loudspeaker outputs are bridge-tied: JAMMA L is not ground. The amplifier stays off until Pico C raises AMP_EN.



Sheet: /audio/
File: audio.kicad_sch

Title: Famicom JAMMA board – Audio: mixer, loop through the cartridge, pow

Size: A3	Date:	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 7/9

JAMMA edge connector, inputs, coin counter and lock-out drivers

Every input: 1 k pull-up to +3V3 (3.3 mA through the switch), 4.7 k in series, 10 nF to ground, then a 74HC165 on +3V3.

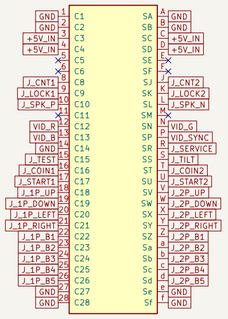
Pico C reads the chain: after a low pulse on IO_LATCH the first bit on IN_DATA is input H of U17, the last is input A of U20.

The same pulse copies the 74HC595's shift register to its outputs, so one cycle reads the inputs and writes the outputs.

Fingers 5/E (−5 V), 6/F (+12 V) and 11/M are not connected. 7/H is the key slot.

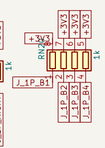
Coin counter and lock-out coils: 5 V or 12 V types, switched to ground; each output has its own clamp, so the board needs no +12 V.

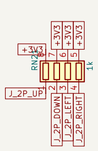
JAMMA edge (gold fingers)

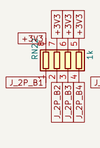


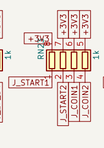
J1

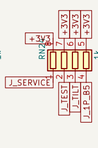


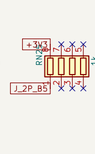


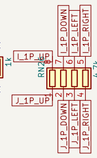


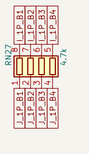


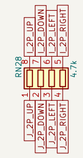


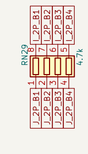


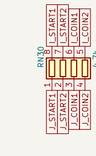


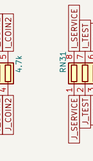


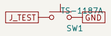


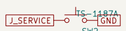


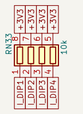


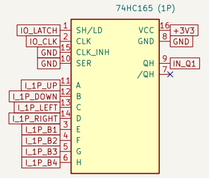




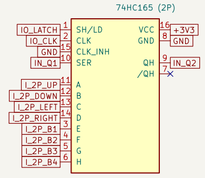




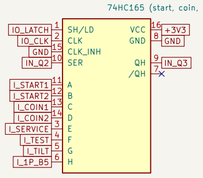




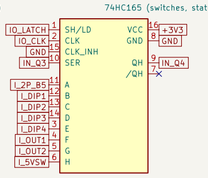
74HC165 (2P)



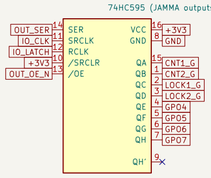
74HC165 (start, coin, service)

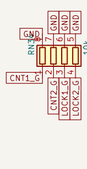


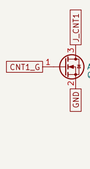
74HC165 (switches, status)

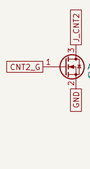


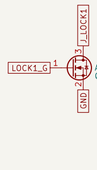
74HC595 (JAMMA outputs)



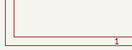


























Sheet: /jamma/
File: jamma.kicad_sch

Title: Famicom JAMMA board — JAMMA edge connector, inputs, coin counter a

Size: A2	Date:	Rev: 0.3
KiCad E.D.A. 10.0.8		Id: 8/9

Three Raspberry Pi Pico 2 modules, USB Type-C socket, links

Pico A (video): clock and reset for CPU and PPU, palette index and sprite bit in, RGB and sync out; the emulator in emulation mode.

Pico B (bus): watches the CPU's accesses to the PPU, replaces palette data on the PPU side, answers the controller reads.

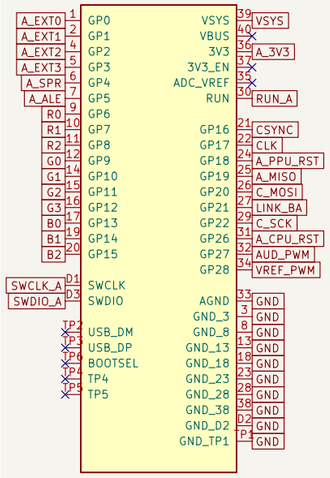
Pico C (input/output and management): JAMMA inputs and outputs, coin and timer logic, cartridge dump, USB Type-C,

power switch of the Famicom half, and reprogramming of Pico A and Pico B through their debug pads.

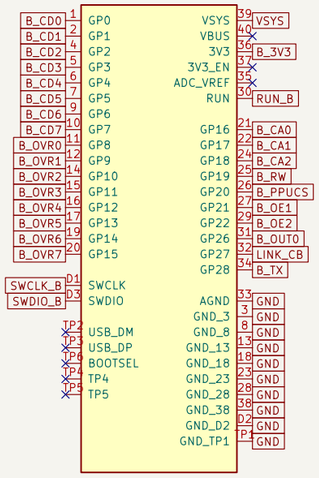
The modules are soldered flat on the board. The USB socket reaches Pico C through the pads under the module (TP2, TP3).

Use either the board's Type-C socket or Pico C's own Micro-USB socket, never both at once: they are the same two wires.

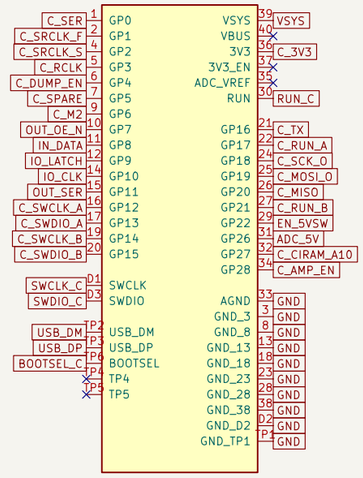
Pico 2 A (video)



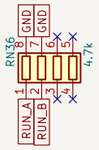
Pico 2 B (bus)



Pico 2 C (I/O, management)









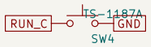


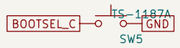




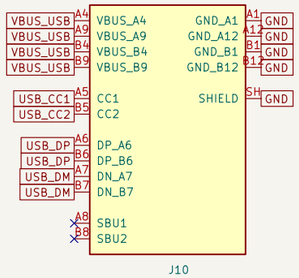








USB Type-C socket (TYPE-C-31-M-12)



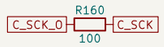




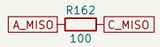










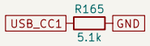


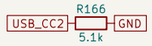




debug probe for Pico C (option)







USB lines (option)



Sheet: /mcu/		
File: mcu.kicad_sch		
Title: Famicom JAMMA board – Three Raspberry Pi Pico 2 modules, USB Type		
Size: A3	Date:	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 9/9